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Unit One 人工智能

1. Reading Material

Passage 1

Tweaking (拧, 扭, 微调) AI software to function like a human brain improves computer's learning ability

Date:

January 12, 2021

Source:

Georgetown University Medical Center

- 1) Computer-based artificial intelligence can function more like human intelligence when programmed to use a much faster technique for learning new objects, say two neuroscientists who designed such a model that was designed to mirror human visual learning.

 **mir·ror** /'mɪə(r)/ /'mɪrər/

VERB ▾

1. ~ sth to have features that are similar to sth else and which show what it is like 反映

SYN REFLECT

⇒ *The music of the time mirrored the feeling of optimism in the country.*  
这个时期的音乐反映出这个国家的乐观精神。

2) In the journal *Frontiers in Computational Neuroscience*, Maximilian Riesenhuber, PhD, professor of neuroscience, at Georgetown University Medical Center,

and Joshua Rule, PhD, a postdoctoral scholar at UC Berkeley, explain how the new approach vastly improves the ability of AI software to quickly learn new visual concepts.

3)"Our model provides a biologically **plausible** way for artificial neural networks to learn new visual concepts from a small number of examples," says Riesenhuber. "We can get computers to learn much better from few examples by **leveraging** prior learning in a way that we think mirrors what the brain is doing."

Corpus

LONGMAN

plausible /'plɔːzəbəl \$ 'plɒː-/ ●○○ *adj* 🔊 🔊

1. reasonable and likely to be true or successful 貌似有理的；似乎是真的

OPP implausible

🔊 His story certainly sounds plausible.
他的叙述听上去当然很合理。

le-ver-age /'li:vərɪdʒ/ /'levərɪdʒ/

VERB ▾

~ sth (*business 商业*) to get as much advantage or profit as possible from sth that you have 充分利用

⇒ The company needs to leverage its resources. 🔊 🔊
该公司需要充分利用其资源。

4)Humans can quickly and accurately learn new visual concepts from sparse data — sometimes just a single example. Even three- to four-month-old babies can easily learn to recognize zebras and distinguish them from cats, horses, and giraffes. But computers typically need to "see" many examples of the same object to know what it is, Riesenhuber explains.

5)The big change needed was in designing software to identify relationships between entire visual categories, instead of trying the more standard approach of identifying an object using only low-level and intermediate information, such as shape and color, Riesenhuber says. "The computational power of the brain's hierarchy lies in the potential to simplify learning by **leveraging** previously learned representations from a databank, as it were, full of concepts about objects," he says.

6) Riesenhuber and Poggio found that artificial neural networks, which represent objects in terms of previously learned concepts, learned new visual concepts significantly faster.

7) Poggio explains, "Rather than learn high-level concepts in terms of low-level visual features, our approach explains them in terms of other high-level concepts. It is like saying that a platypus looks a bit like a duck, a beaver, and a sea otter."

8) The brain architecture underlying human visual concept learning builds on the neural networks involved in object recognition. The anterior temporal lobe of the brain is thought to contain "abstract" concept representations that go beyond shape. These complex neural hierarchies for visual recognition allow humans to learn new tasks and, crucially, **leverage** prior learning.

9) "By reusing these concepts, you can more easily learn new concepts, new meaning, such as the fact that a zebra is simply a horse of a different stripe," Riesenhuber says.

10) Despite advances in AI, the human visual system is still the gold standard in terms of ability to **generalize** from few examples, robustly deal with image variations, and comprehend scenes, the scientists say.

gen·er·al·ize (BrE also gen·er·al·ise) /'dʒenrəlaɪz/ /'dʒenrəlaɪz/

VERB ▾

1. [I] ~ (from sth) to use a particular set of facts or ideas in order to form an opinion that is considered valid for a different situation 概括; 归纳

11) "Our findings not only suggest techniques that could help computers learn more quickly and efficiently, they can also lead to improved neuroscience experiments aimed at understanding how people learn so quickly, which is not yet well understood," Riesenhuber concludes.

This work was supported in part by Lawrence Livermore National Laboratory and by the National Science Foundation (1026934 and 1232530) Graduate Research Fellowship Grants.

(<https://www.sciencedaily.com/releases/2021/01/210112085359.htm>)

Notes

1) Neuroscience: Neuroscience is an interdisciplinary science that works closely with other disciplines, such as mathematics, linguistics, engineering, computer science, chemistry, philosophy, psychology, and medicine. Neuroscientists study the cellular, functional, behavioral, evolutionary, computational, molecular, and medical aspects of the nervous system. There are various fields that focus on different aspects, but they often overlap.

2) Leverage (Para.3): ['levərɪdʒ] 杠杆, 举债经营, 补充, 对...产生影响, 利用

3) Platypus (para.7): ['plætɪpʊs] 鸭嘴兽

4) Sea otter (para.7): [si:'ɑ:tər] 海獭; 水獭

5) anterior temporal lobe (para.8): [æn'tɪrɪər 'tempərəl loʊb] 颞叶前部

Passage 2

Building machines that better understand human goals

A new algorithm capable of **inferring** goals and plans could help machines better adapt to the imperfect nature of human planning.

infer /ɪnˈfɜː ʒ - 3: ɪ r/ ●○○ (AWL) **verb** (inferred, inferring) [T] 🔊 🔊

to form an opinion that something is probably true because of information that you have [根据资料] 推断, 推定

infer something from something

🔊 A lot can be inferred from these statistics.

根据这些统计资料可以推断出很多事情。

Rachel Gordon | MIT CSAIL

Publication Date: December 14, 2020

1) While there's been considerable work on **inferring** the goals and desires of agents, much of this work has assumed that agents act optimally to achieve their goals.

2) However, the team was particularly inspired by a common way of human planning that's largely sub-optimal: not to plan everything out in advance, but rather to form only partial plans, execute them, and then plan again from there. While this can lead to mistakes from not thinking enough "ahead of time," it also reduces the cognitive load.

3) For example, imagine you're watching your friend prepare food, and you would like to help by figuring out what they're cooking. You guess the next few steps your friend might take: maybe preheating the oven, then making dough for an apple pie. You then "keep" only the partial plans that remain **consistent** with what your friend actually does, and then you repeat the process by planning ahead just a few steps from there.

consistent /kənˈsɪstənt/ ●●○ (S3) (W3) (AWL) **adj** 🔊 🔊

1. always behaving in the same way or having the same attitudes, standards etc – usually used to show approval [行为、态度、标准等] 一贯的, 一致的 [一般含褒义] **OPP** inconsistent

🔊 She's the team's most consistent player.

她是该队中表现最为稳定的选手。

be consistent with something

if a fact, idea etc is consistent with another one, it seems to match it

与某事物相一致，与某事物相吻合

Her injuries are consistent with having fallen from the building.

她的伤和从大楼坠落的情况是吻合的。

The results are consistent with earlier research.

结果与先前的研究相符合。

4)Once you've seen your friend make the dough, you can restrict the possibilities only to baked goods, and guess that they might slice apples next, or get some pecans for a pie mix. Eventually, you'll have eliminated all the plans for dishes that your friend couldn't possibly be making, keeping only the possible plans (i.e., pie recipes). Once you're sure enough which dish it is, you can offer to help.

5)The team's inference algorithm, called "Sequential Inverse Plan Search (SIPS)", follows this sequence to infer an agent's goals, as it only makes partial plans at each step, and cuts unlikely plans early on. Since the model only plans a few steps ahead each time, it also accounts for the possibility that the agent — your friend — might be doing the same. This includes the possibility of mistakes due to limited planning, such as not realizing you might need two hands free before opening the refrigerator. By detecting these potential failures in advance, the team hopes the model could be used by machines to better offer assistance.

6)"One of our early insights was that if you want to infer someone's goals, you don't need to think further ahead than they do. We realized this could be used not just to speed up goal inference, but also to infer intended goals from actions that are too shortsighted to succeed, leading us to shift from scaling up algorithms to exploring ways to resolve more fundamental limitations of current AI systems," says Vikash Mansinghka, a principal research scientist at MIT and one of Tan Zhi-Xuan's co-advisors, along with Joshua Tenenbaum, MIT professor in brain and cognitive sciences. "This is part of our larger moonshot — to reverse-engineer 18-month-old human common sense."

Moonshot (para.6): 一个疯狂的想法或不大可能实现的项目

reverse engineering *noun*

a situation in which a product is examined to see how it is made, so that it can be copied 反向工程, 逆向工程, 倒序制造 [指研究某种产品的结构继而进行仿制]

7)The work builds conceptually on earlier cognitive models from Tenenbaum's group, showing how simpler inferences that children and even 10-month-old infants make about others' goals can be modeled quantitatively as a form of Bayesian inverse planning.

8)While to date the researchers have explored inference only in relatively small planning problems over fixed sets of goals, through future work they plan to explore richer hierarchies of human goals and plans. By encoding or learning these hierarchies, machines might be able to infer a much wider variety of goals, as well as the deeper purposes they serve.

9)“Though this work represents only a small initial step, my hope is that this research will lay some of the philosophical and conceptual groundwork necessary to build machines that truly understand human goals, plans and values,” says Xuan. “This basic approach of modeling humans as imperfect reasoners feels very **promising**. It now allows us to infer when plans are mistaken, and perhaps it will eventually allow us to infer when people hold mistaken beliefs, assumptions, and guiding principles as well.”

promising /'prɒməsɪŋ \$ 'praɪ-/ ●○○ *adj*

showing signs of being successful or good in the future 大有希望的, 很有前途的

10)Zhi-Xuan, Mansinghka, and Tenenbaum wrote the paper alongside EECS graduate student Jordyn Mann and PhD student Tom Silver. They virtually presented their work last week at the Conference on Neural Information Processing Systems (NeurIPS 2020).

(<https://news.mit.edu/2020/building-machines-better-understand-human-goals-1214>)

Notes:

1) In the quest to capture the social intelligence in machines, researchers from MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL) and the Department of Brain and Cognitive Sciences created an algorithm capable of inferring goals and plans, even when those plans might fail. To create their model the team used Gen, a new AI programming platform recently developed at MIT, to combine symbolic AI planning with Bayesian inference. The team's model performed 20 to 150 times faster than an existing baseline method called Bayesian Inverse Reinforcement Learning (BIRL), which learns an agent's objectives, values, or rewards by observing its behavior, and attempts to compute full policies or plans in advance.

2) agent:具有自主性、交互性等特征的活着的计算实体。

3) Moonshot (para.6): 一个疯狂的想法或不大可能实现的项目

4) Reverse engineer (para.6): Reverse engineering is taking apart an object to see how it works in order to duplicate or enhance the object. (<https://baike.baidu.com>)

5) Bayesian inverse planning(para.7): a model of rational planning. Inverse planning refers to the problem of learning a probabilistic model of the dynamics of a robot (or a human) by observing her optimal behavior performing a given task.

Passage 1

The business of quantum computers

Date: September 26, 2020 Source: *The Economist* P72-73

Quantum computing is attracting commercial interest.

Big, stable quantum computers would be useful devices. By exploiting the counter-intuitive properties of quantum mechanics, they could perform some calculations (though only some) faster than any **conceivable** non-quantum machine.

conceivable /kən'si:vəbəl/ ●○○ (AWL) *adj* 🔊 🔊

able to be believed or imagined 可以相信的; 可想象的 (OPP) **inconceivable**

For one thing, they would probably be much more rapid than any classical computer at searching a database—an elemental operation with a thousand uses. They would be quicker at more specific tasks, too. Precisely simulating all but the simplest chemical reactions is mathematically **intractable** for any non-quantum computer, no matter how huge.

intractable /ɪn'træktəbəl/ *adj* [FORMAL] 🔊 🔊

1. an intractable problem is very difficult to deal with or solve 〔问题〕难对付的, 难解决的

A quantum computer could do this, aiding the development of drugs, catalysts and batteries. Quantum computers would also speed up the analysis of optimisation problems, which try to find the best way to **juggle** many variables

2. [I, T] to try to fit two or more jobs, activities etc into your life, especially with difficulty 〔尤指吃力地〕同时应付〔几份工作、多项活动等〕

in order to maximise a particular outcome. That would be a **boon** to the

boon /bu:n/ *noun* [C USUALLY SINGULAR] 🔊 🔊

something that is very useful and makes your life a lot easier or better 裨益; 为生活带来方便的东西; 有用之物

transport industry (by finding efficient routes) and to finance (by maximising profits given a set of constraints). Boston Consulting Group (BCG), a management consultancy, foresees quantum computers improving the operating income of their users by between \$450bn and \$850bn a year by 2050.

Unfortunately, big, stable quantum computers do not yet exist. But small,

ornery, unstable ones do. 【John Preskill, a quantum-computing researcher at

ornery /'ɔ:nəri \$ 'ɔ:r-/ *adj* [AmE] 🔊 🔊

behaving in an unreasonable and often angry way, especially by doing the opposite of what people want you to do 脾气坏的, 执拗的

🔊 | an ornery kid
一个坏脾气的小孩

the California Institute of Technology, **dubs** [2021/2022 年期末考试] such machines NISQS—Noisy, Intermediate-Scale Quantum

dub¹ /dʌb/ *verb* (dubbed, dubbing) [T] 🔊 🔊

1. to give something or someone a name that describes them in some way 把...称为, 给...起绰号 → **label, name**
computers.】

Some see NISQS as mere stepping stones towards size and stability, and that is certainly the goal of those working on them. A growing number of companies and investors, however, are hopeful that NISQS themselves will be able to do useful work in the meantime. These firms are hunting for "quantum advantage"—a way in which even today's limited machines might have an impact on their bottom lines, or those of their customers.

Let a hundred flowers bloom

For most of the field's history, quantum-computing research has been backed by governments or big information-technology firms. Increasingly, though, the venture-capital (VC) industry is showing an interest. PitchBook, a research firm based in Seattle, has tracked \$495m of vc money that has been invested in quantum computing so far this year—almost double last year's total. Dozens of **startups** 【创业企业】 are competing with the **incumbent** 【现任的】 tech giants. And **established** 【资深的】 companies, such as Daimler, a carmaker, and Goldman Sachs, a bank, are beginning to experiment with the **nascent** 【新生的】 industry's products, hoping that, if they can master them, they will bestow an advantage over their competitors.

Much of the money is going towards building hardware. Doug Finke, a physicist who runs the Quantum Computing Report, which focuses on the nascent industry, is aware of 87 organisations, big and small, trying to construct quantum computers. Unlike classical computing, which had settled by the 1970s on silicon transistors as the units of computation, there is, as yet, no consensus on the best way to build a quantum computer. IonQ, a firm in Maryland that has raised \$84m, uses trapped ytterbium ions, manipulated by lasers, to perform its calculations. Rigetti Computing, a Californian company which announced earlier this month that it would be building a quantum computer for the British government, Rigetti Computing employs microwaves to control pairs of electrons flowing through superconducting circuits. Microsoft, although very much not a startup, is working on a

"topological" quantum computer that relies on the interactions of super-cold electrons.

One particularly well-financed new firm is PsiQuantum, which does its computing with photons that run along waveguides **etched** 【蚀刻; 凿刻】 onto ordinary silicon chips. It hopes to leapfrog the NISQ era entirely and produce a fully-fledged quantum computer within about five years—though that is far sooner than most experts think **feasible**.

feasible /'fi:zəbəl/ ●○○ *adj* 🔊 🔊

a plan, idea, or method that is feasible is possible and is likely to work

〔计划、想法或方法〕可行的, 可实行的, 行得通的 **SYN** possible

■ PsiQuantum was founded in 2015 by researchers from Imperial College and the University of Bristol, both in Britain. It has raised \$215m from backers including Black-Rock, a giant investment firm, and the Founders Fund, an American vc company that was an early investor in Space X and Facebook.

...Microsoft (whose own topological machines are still at an early stage of development) offers access via Azure, its cloud-computing service, to machines from IonQ, Honeywell and a company called Quantum Circuits. Amazon hosts machines from Rigetti, IonQ and D-Wave, a Canadian company that builds specialised, fixed-purpose computers called quantum annealers. Rather than have hardware-makers come up with useful algorithms, the hope is that existing businesses, by being given access to the machines in this way, will invent such algorithms for themselves. With that in mind, says Krysta Svore, who leads Microsoft's quantum-computing research, one task is to make those machines as easy as possible to use. The firm has developed

tools intended to help customers wrap their brains around the counter-intuitive properties of quantum computers. It has also developed software intended to **smooth over**【消除；缓和】 differences between machines from different makers, so that programmers do not have to worry about whether they are marshalling ions or electrons. She compares this to the early days of classical computing, when the development of compilers and early programming languages freed human beings from the need to think in the ones and zeros of raw machine code.

The big question is what all this is leading up to. Despite the excitement, the industry's commercial pioneers have their work cut out for them. There is plenty of promise, but, as yet, no certainty. Finding algorithms that are both commercially useful and simple enough to work within a NISQ machine's limitations is not easy. A report published last year by America's National Academy of Sciences reminded readers that no commercial applications are currently known to exist.

The first step, then, is to go looking. The industry has been **cheered**【(为...) 欢呼，喝彩；(为...) 加油】 by Google's demonstration last year of "quantum supremacy", in which it used a NISQ machine to perform, with minutes of computing time, a calculation that would have taken thousands of years on classical hardware. Google's calculation was highly **contrived** and of little use in the real world. But it proved a point. ...

con·trived /kən'traɪvd/ /kən'traɪvd/

ADJECTIVE

(disapproving)

planned in advance and not natural or genuine; written or arranged in a way that is not natural or realistic 预谋的；不自然的；人为的；做作的

Passage 2

ByteDance

China's first global software company is **going from strength to strength**. America

go from strength to strength

doesn't like it one bit. to become more and more successful 不断取得成功; 日益壮大

【doesn't like it one bit 一个俚语, I don't like that idea one bit=I hate that idea, a bit 可以解释为 a (small) piece, part, segment, slice; moment, while】

As covid-19 has forced the world's teenagers out of school and into their rooms, they have turned to a familiar digital companion, TikTok. The short-video app was downloaded 115m times in March. Its nearly 1bn regular users enjoy silly clips of dog **antics** alongside pandemic advice from the World Health Organisation. Collectively, TikTok videos tagged with #coronavirus have been watched 53bn times.

antics /'æntɪks/ **noun** [PLURAL]  

behaviour that seems strange, funny, silly, or annoying 古怪可笑的举动

 We're all growing tired of his childish antics.
我们都厌倦了他那些幼稚的滑稽举动。

► see thesaurus at **behaviour**

TikTok's popularity over the past two years has **shone the spotlight on** [shine the spotlight 直译为点亮.....照亮, 常用指聚焦于 XX, XX 备受关注、备受瞩目]ByteDance, its Beijing-based developer. Founded by a Chinese computer scientist, Zhang Yiming, in 2011, it is now the world's biggest unlisted technology "unicorn", recently valued at between \$90bn and \$100bn. It is also the only technology firm **bar** bar 这里是“除了.....之外”的意思 Apple with more than 100m users in both China and America, where TikTok has taken on the likes of YouTube and Instagram.

And ByteDance isn't done. The 60,000 people in its buzzy offices—"We are like flies," says one former employee—**crank out**[快速地大量制造; (尤指) 粗制滥造] one app after another. In the past year it has launched a worldwide corporate-software service (Lark), a music-streaming app in India and Indonesia (Resso) and, in China, a messaging rival to

WeChat. As other firms **sack** [解雇] workers amid covid-19, ByteDance is hiring 10,000 globally. It plans to employ 30,000 on top of that this year.

ByteDance is not the first Chinese firm with foreign ambitions. Commodity giants such as CNOOC, an oil firm, have been buying foreign reserves, and rivals, since the 1990s. In the past decade industrial giants have pursued Western competitors from car making (as with Geely's purchase of Volvo) to chemicals (ChemChina's of Syngenta). More haphazardly, conglomerates like Fosun and Anbang **splurged on** trophy assets (including ClubMed and the Waldorf Astoria hotel, respectively).

3. splurge on 挥霍; 浪费

splurge sth on sth 花比往常更多或超过能力的钱用在某东西上, 同义词 piss away, to squander.

Unlike most of its **peripatetic** [巡回工作的; 流动的] predecessors, though, ByteDance has built its empire by making products that **appeal** [有吸引力] beyond China. It is China's first global software giant. It has also **courted** [奉承, 讨好] foreign investors. Filings in Hong Kong show that it has used an arrangement called the "Sina structure", which allows it to accept money from abroad, since its founding (many Chinese tech firms turn to this structure only when eyeing a stock market listing in America). Around 80% of ByteDance's investors are non-Chinese. So are four of its five board members. The other is Mr. Zhang.

All this helps ensure ByteDance is not, in contrast to many Chinese tech upstarts, **in**

hock to Alibaba,

in hock
[INFORMAL]
a) in debt 负债
be in hock to somebody
The fashion chain is still in hock to the banks.
那家时装连锁店仍欠着银行的债。

Baidu and Tencent—which is handy, for

ByteDance competes with China's tech titans for ad revenue. It makes most of its money selling ads in its two main Chinese apps: Douyin, a Chinese TikTok, and Toutiao, a multimedia-and-news app akin to Facebook's news feed (its WeChat rival, Duoshan, disappointed).

How much money it makes, exactly, is unclear: as a private company ByteDance does not publish its accounts. But leaks and statements from investors put last year's revenue at between 104bn yuan and 140bn yuan (\$15bn-20bn), more than Uber, Snapchat and Twitter combined. Its ad revenues in China **surpassed** [超越] Tencent's and Baidu's and now **trail**

[落后于]only those of Alibaba. It reportedly turned a profit in June 2019, a **feat** in the world of loss-making unicorns.

If the firm generates \$25bn in sales this year, as it is expected to despite covid-19, it will have done so three years faster than Facebook. It is the only big Chinese firm whose share of the domestic advertising market is growing fast, from 9% to 17% in 2019, according to Bernstein, a research firm. A recently launched advertising network, Pangle, which lets advertisers reach consumers across any of its non-Chinese apps, may at last help it monetise TikTok.

No wonder American rivals are trying to **clip its wings**. [限制某人的权力或影响力；约束或抑制某人] Facebook's TikTok clone, Lasso, has **flopped**.

flop¹ /flɒp \$ flɑ:p/ **verb** (flopped, flopping)

INFORMAL if something such as a product, play, or idea flops, it is not successful because people do not like it [产品、戏剧、想法等因无人喜欢而] 失败，砸锅

YouTube is said to be developing another (called Shorts). But the biggest threat has emerged in the form of America's government, **wary** of [害怕] China's rise in any sphere.

Politicians in Washington fear that data on American users is being handed to Beijing; that Chinese algorithm designers are infecting **impressionable** ([尤因年轻而] 易受影响的) Western youngsters with communist propaganda; and that ByteDance content is censored in line with party **whims** (突发的念头，一时的兴致 [尤指没有重要或充分的理由时]) . In March two Republican senators introduced a bill that would ban TikTok from all government devices. The Committee on Foreign Investment in the United States (CFIUS) is reviewing ByteDance's acquisition in 2018 of Musical.ly, another app, which had 60m users in America and Europe.

In statements, TikTok has said that it stores all American "user data in the United States" and that its American operations are "not influenced by any foreign government, including the Chinese government". The firm is trying to **mollify** [抚慰，使平静，使平息怒气] critics and has hired prominent security and legal experts in America to help it. Last month it opened a "transparency centre" in Los Angeles, where it promises to share

information about its content-moderation, privacy and security controls, and said it would stop using moderators in China to handle content from users outside the country.

That, most lawyers and technologists reckon, may help get CFIUS off its back. Listing ByteDance in Hong Kong, which is **under consideration**【在考虑中；在考虑之中】, may also **foster**【促进，培养，助长】 trust. So would **anointing**【指定】 a non-Chinese successor—though, at 37, Mr Zhang may not be ready to retire. As a last resort, he is said to be willing to contemplate **spinning** TikTok **off**【spin off 分拆；剥离】. That would **cement**【加强，巩固（关系、协定等）】 the digital divide between China and the West—until another hit ByteDance app tries to bridge it.

Unit Three

Reading Materials

Passage 1

The Transformative Power of 5G

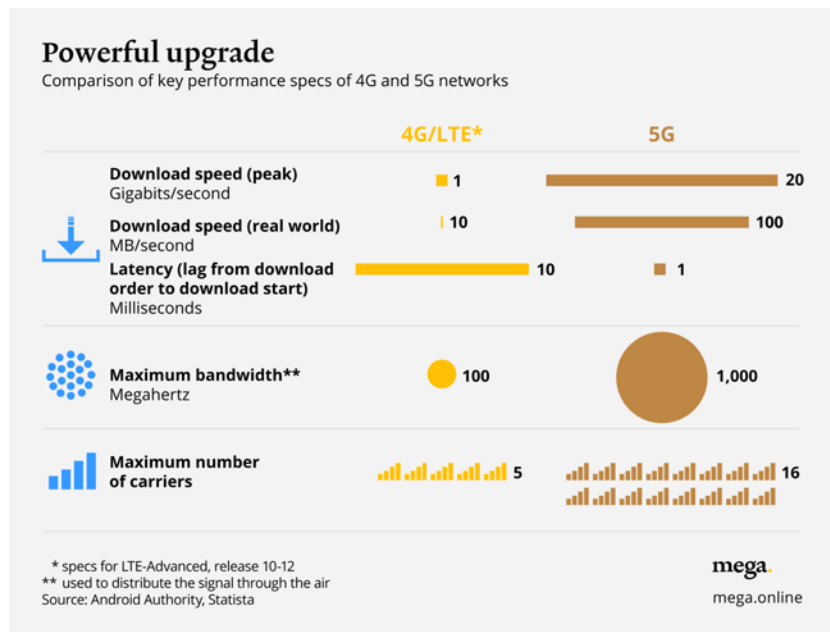
This once-in-a-decade mobile phone network upgrade is expected to place us **squarely** 【直接地；坚定地】 in the middle of the **Internet of Things**.

by Scientific American Custom Media

1 Major **carriers** 【通信公司】 are in the process of rolling out ultra-high-speed mobile-phone networks that use fifth-generation (5G) technology, **transmitting** 【传输，发送，传送】 data up to 50 times faster than current phone networks. But the promise of 5G is that it will provide more than just a **boost** 【促进，推动，使兴旺】 to our phone service and download speeds. “5G will be one of the more transformative technologies we’ve had so far,” says Karri Kuoppamaki, vice president of radio network technology and strategy at **T-Mobile**, where he is overseeing the company’s rollout of 5G networks.

2 There are three differences between 5G and the current 4G: speed, bandwidth and what network specialists call latency, which is the time it takes for data to move from a server on the network to a device. 5G networks are faster, can handle a higher volume of data, and reduce wait times. The new technology is able to shoot data from millions of local sensors or gadgets to distant cloud computers for processing and then return the relevant guidance — all in a fraction of a second. This **deluge** 【纷至沓来的事物(如信件、问题等)】 of data **flooding through** 【涌入】 our society will enable a host of new services, including connecting new devices and equipment.

3 Sensors themselves can be tinier, cheaper and more durable because of yet another advantage of 5G: smaller batteries. 5G devices stay connected to the network instead of engaging in the back-and-forth signaling that 4G networks require for connectivity, which **saps** battery life. 5G will power up our daily lives. “We’ll constantly have the information we need to make decisions,” Kuoppamaki predicts.



4 The first 5G networks came online in late 2019 in major cities around the world. That’s more or less in keeping with the pattern of major decadal upgrades in phone-network capabilities. To get a sense of how influential these upgrades can be, consider that the last big upgrade — from 3G to 4G networks around 2010 — drove the **proliferation** 【激增】 of phone apps that are now integral to our connected culture. “The move to 4G created the app economy, and made Google and Facebook what they are today,” says Kuoppamaki.

5 The first applications to benefit from 5G will likely be those involving **live-streaming video** 【直播视频】 — that is, using a phone to record and transmit an event as it is happening. 5G speeds can cope with **8K image resolution** 【分辨率】, which **entails** 【牵涉; 需要】 quadruple the

amount of information of **ultra-high-definition 4K images** and beyond even the capabilities of most TVs. The network also has enough extra bandwidth to simultaneously transmit multiple other video streams, allowing users to switch between camera angles at sporting events, for example, or to **pan** 【〔拍摄电影或电视时摄影机〕摇摄，摇动拍摄】 the viewpoint around 360 degrees.

6 Other applications certain to get early 5G upgrades include mobile gaming and **virtual reality (VR)** apps that embed goggle-wearing users in 3D images. These applications will benefit from another aspect of 5G networks. “There’s a **misconception** 【误解】 that 5G is only about higher speeds,” says Kuoppamaki. “Low latency is a big benefit, too, a more comprehensive, end-to-end measure of data speed than transmission speed.” Gaming and VR apps work best when the images change almost instantly in response to user input, whether it’s swiping on the screen, waving a hand or turning a head. The new 5G networks will reduce latency by 80 percent, providing a near immediate response.

7 The speed and fast response of 5G networks will also affect the kit: **VR goggles**, for example, can be **sleeker** 【sleek 线条流畅的，造型优美的】，lighter and less clunky, notes Kuoppamaki. That’s because today those goggles need to provide much of the processing power that turns raw data into actual images. But 5G networks can zip data back and forth quickly enough to mean that much of the processing can move to the cloud.

8 Closely related to VR is **augmented reality (AR)**, where graphics and text are **superimposed** 【使〔图画、图像、照片〕重叠，使叠加】 on real images. AR, too, will become widespread with 5G, and in many cases essential, says Kuoppamaki. “Imagine you want to try a little home plumbing to change your faucet,” he says. “You could put on your goggles and let an augmented-reality app walk you through each step under the sink, while you look at your pipes.” 5G-charged VR and

AR apps could literally add a new dimension to education and training in almost any domain.

9 And, of course, there is the Internet of Things, which will surround us with many ‘smart’ items. Around the home, that could be tomato plants in the garden that can ask for water, roofs that warn of weaknesses after extreme weather, jackets that keep parents updated on a child’s location, trash cans that ask to be emptied, and milk cartons that point out expiration dates. In the streets, it’s a heads-up on open parking spaces, second-to-second data on surrounding traffic for driverless cars, local air-quality warnings, icy sidewalk alerts, and much more.

10 “The Internet of Things will change the way we live and how we interact with the environment around us,” says Kuoppamaki.

11 Businesses will benefit, too. Industrial robots, drones and driverless trucks will be more responsive to obstacles and fast-changing conditions. Cables will be redundant, even for the most complex computer-controlled machines, making them easier to move around. Businesses will be free to reconfigure the factory floor in response to product changes or fluctuations in customer orders. “Today, it’s hard for factories to change quickly,” says Kuoppamaki. “If you can make equipment more mobile, you can reduce the time to market.” What’s more, he adds, sensors in the machinery will constantly feed remote AI software with the monitoring data it needs to self-adjust in response to variations in product quality, and to judge when the equipment requires preventive maintenance to avoid **down-time** 【停工；停止运行时间】 and more expensive repairs.

12 The 5G transformation won’t happen overnight, warns Kuoppamaki. The first wave of sophisticated applications are likely three or more years away. Cost may be a temporary barrier, too because currently, the only phones that can access 5G networks are top-of-the-line new

models.

13 But the initial benefits, mostly those relating to speeding up existing applications, are already arriving.

14 “Like all new technologies, adoption starts in the high end, and then makes its way down,” he says. “It will only take a few years before we see 5G everywhere.”

<https://www.scientificamerican.com/custom-media/pictet/the-transformative-power-of-5g/>

Notes

1) Internet of Things (IOT) 物联网

A system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

2) T-Mobile

A multinational mobile phone operator. It is a subsidiary of German Telecom and belongs to the Freemove Union. T-Mobile operate GSM networks in Western Europe and the United States and participate in network operations in Eastern Europe and South-East Asia through financial means. The company has 109 million users and is one of the world's largest mobile phone companies.

3) 8K image resolution 8K 分辨率

4) ultra-high-definition 4K images 超高清 4K 图像

5) virtual reality (VR) 虚拟现实

An artificial environment created with computer hardware and software and presented to the user in such a way that it appears and feels like a real environment.

6) VR goggles VR 眼镜即 VR 头显, 虚拟现实头戴式显示设备

7) augmented reality (AR) 增强现实

A type of virtual reality that aims to duplicate the world's environment in a computer. An augmented reality system generates a composite view for the user that is the combination of the real scene viewed by the user and a virtual scene generated by the computer that augments the scene with additional information.

8) down-time 停机时间; 停工期

Passage 2

Qualcomm makes strides in 5G network development: Here's what it means for you

By [Christian de Looper](#) October 21, 2020

1 **Qualcomm** wants to make 5G more **accessible**【易到达的; 易进入的】, more common, and faster. The company recently held its annual 5G Day event, where it made a few announcements related to the rollout of 5G, the continued development of 5G tech, and more.

2 Of course, you might be wondering whether any of this really matters to you and how it will benefit the average smartphone user. Here's a quick **rundown** 【简要报告, 扼要说明, 梗概】 of Qualcomm's new 5G-related announcements, and what they could mean for you.

Qualcomm wants to help build 5G infrastructure

3 Traditionally, Qualcomm has developed and manufactured chipsets and modems that can connect to 5G networks, but now it wants to play a bigger role in the actual infrastructure behind 5G. Specifically, the company is building **Radio Access Network (RAN)** chips, which are designed to be used in 5G base stations for your phone (or tablet, or computer) to connect.

4 It's important to note that Qualcomm won't necessarily be directly competing with the likes of Huawei and Ericsson. Instead, it will supply those companies with the 5G chips and related software to power their infrastructure. That marks a shift for cell infrastructure, which has traditionally been built all by one company. Making 5G networks a little more modular also means that they could be more easily updated and developed without carriers having to deploy

all-new infrastructure, but it remains to be seen if that will actually happen.

5 But what does that mean for you? Well, Qualcomm has been a leader in the 5G world for some time, and its entering this particular business could help make real-world 5G networks faster and more reliable. Unlike other companies, Qualcomm has expertise in the end-user device chipsets, too, so it knows exactly how to make them **interoperate** 【交互操作】 for the best performance. Qualcomm's chips, of course, support the super-fast **mmWave (millimeter-wave)** networks, and the widespread **Sub-6** networks, so if they become widespread, you may get much better mmWave support.

There's a new record for fastest 5G speeds

6 Qualcomm also announced that in partnership with **Verizon** and **Ericsson**, it achieved a peak 5G speed of a whopping 5.06Gbps. The speed was achieved with a bunch of fancy tech on a mmWave network and using what's called **carrier aggregation**, which combines multiple channels to make for faster speeds.

7 According to Qualcomm, 5G will eventually reach speeds of 10Gbps, but that's just talk right now — we'll have to wait and see if that happens and if it's accessible.

8 It's important to be very clear about how Qualcomm achieved this huge peak speed — in a tightly controlled lab environment. In other words, carriers aren't going to be able to flick a switch and turn on 5Gbps speeds anytime soon.

9 But that doesn't mean that it won't benefit you. As companies like Qualcomm continue to work on 5G development, and as the tech gets better and better it will eventually make its way to

the real world. When it does, you likely won't get 5Gbps, considering things like distance, obstacles, and the number of people using the same cell tower as you. But you should still get faster speeds than you did before, and hopefully, faster speeds than you'll get on even the best 4G **LTE** networks.

<https://www.digitaltrends.com/mobile/qualcomm-5g-day-2020-roundup/>

Notes

1) Qualcomm 高通公司

Founded in 1985 and headquartered in San Diego, California, USA, Qualcomm is the world's leading wireless technology innovator and has changed the way the world connects, computes and communicates. Its invention opened the era of mobile interconnection.

2) Radio Access Network (RAN) 无线电接入网

A part of a mobile communication system and the realization of radio access technology. It exists on a device, a mobile phone, a computer, or any remotely controlled machine with a core network and provides a communication connection between the two.

3) mmWave (millimeter-wave)毫米波

Millimeter wave is an electromagnetic wave between microwave and light wave. Usually millimeter wave band refers to 30 GHz~300 GHz, corresponding wavelength is 1 mm~10 mm. The millimeter wave communication refers to the communication with millimeter wave as the carrier of information transmission.

4) Sub-6

The global 5 G network frequency band is mainly divided into two major ranges: Sub-6GHz and millimeter wave. At present, the main use of this band in China is Sub-6GHz, which has strong signal penetration but slower speed of data transmission than that of millimeter wave.

5) Verizon 威瑞森电信

The largest local telephone company and the largest wireless communications company in the United States.

6) Ericsson 爱立信公司

Founded in 1876 in Stockholm, Sweden., Ericsson is the world's largest provider of mobile communications networks.

7) carrier aggregation 载波聚合

The aggregation of two or more carrier units (Component Carrier) to support a larger transmission bandwidth (up to 100 MHz).

8) LTE (Long Term Evolution)长期演进技术

LTE (Long Term Evolution) project is an evolution of 3G, a transition between 3 G and 4 technologies, and a global standard of 3.9 G, which improves and enhances 3G air access technology by using OFDM and MIMO as the sole criteria for its wireless network evolution.

5. prosthetics: 弥补术; 修复术

6. caution: v. 警告; 劝告

7. exoskeleton ['eksouˌskelɪtn]: 外骨骼; 皮骨骼

8. respirocyte: 呼吸细胞

9. dub: 把...叫做

10. homo robocopus: 人机结合体

11. be reduced: 使变成

12. bionic: 仿生学的

Comprehension questions:

1. How does France seek to enhance the cognitive, perceptive, psychological and physical capabilities of its troops?
2. What is biased attitude the report takes to China's development of soldiers with biologically enhanced capabilities?
3. What is potential risk in developing bionic men?

Unit 4

Passage 2

Reading Material (982 words):

Designing customized “brains” for robots

A new system devises hardware architectures to hasten robots’ response time

Daniel Ackerman | MIT News Office

Publication Date: January 21, 2021

Contemporary robots *can* move quickly. “The motors are fast, and they’re powerful,” says Sabrina Neuman.

Yet in complex situations, like interactions with people, robots often *don’t* move quickly. “The ~~hang up~~ is what’s going on in the robot’s head,” she adds.

Perceiving ~~stimuli~~ and calculating a response takes a “~~boatload~~ of computation,” which limits reaction time, says Neuman, who recently graduated with a PhD from the MIT Computer Science and Artificial Intelligence Laboratory (CSAIL). Neuman has found a way to fight this mismatch between a robot’s “mind” and body. The method, called ~~robomorphic~~ computing, uses a robot’s physical layout and intended applications to generate a customized computer chip that minimizes the robot’s response time.

The advance could fuel a variety of robotics applications, including, potentially, frontline medical care of contagious patients. “It would be fantastic if we could have robots that could help reduce risk for patients and hospital workers,” says Neuman.

Neuman will present the research at this April's International Conference on Architectural Support for Programming Languages and Operating Systems. MIT co-authors include graduate student Thomas Bourgeat and Srini Devadas, the Edwin Sibley Webster Professor of Electrical Engineering and Neuman's PhD advisor. Other co-authors include Brian Plancher, Thierry Tambe, and Vijay Janapa Reddi, all of Harvard University. Neuman is now a postdoctoral NSF Computing Innovation **Fellow** at Harvard's School of Engineering and Applied Sciences.

There are three main steps in a robot's operation, according to Neuman. The first is perception, which includes gathering data using sensors or cameras. The second is mapping and localization: "Based on what they've seen, they have to construct a map of the world around them and then localize themselves within that map," says Neuman. The third step is motion planning and control — in other words, plotting a course of action.

These steps can take time and an awful lot of computing power. "For robots to be deployed 【部署, 调度】 into the field and safely operate in dynamic environments around humans, they need to be able to think and react very quickly," says Plancher. "Current **algorithms** cannot be run on current CPU hardware fast enough."

Neuman adds that researchers have been investigating better algorithms, but she thinks software improvements alone aren't the answer. "What's relatively new is the idea that you might also explore better hardware." That means moving beyond a **standard-issue** CPU processing chip that comprises a robot's brain — with the help of hardware acceleration.

Hardware acceleration refers to the use of a specialized hardware unit to perform certain computing tasks more efficiently. A commonly used hardware accelerator is the graphics processing unit (GPU), a chip specialized for parallel processing. These devices are handy for graphics because their parallel structure allows them to simultaneously process thousands of pixels. “A GPU is not the best at everything, but it’s the best at what it’s built for,” says Neuman. “You get higher performance for a particular application.” Most robots are designed with an intended set of applications and could therefore benefit from hardware acceleration. That’s why Neuman’s team developed robomorphic computing.

The system creates a customized hardware design to best serve a particular robot’s computing needs. The user inputs the parameters of a robot, like its limb layout and how its various joints can move. Neuman’s system translates these physical properties into mathematical matrices. These matrices are “sparse,” meaning they contain many zero values that roughly correspond to movements that are impossible given a robot’s particular anatomy. (Similarly, your arm’s movements are limited because it can only bend at certain joints — it’s not an infinitely pliable spaghetti noodle.)

The system then designs a hardware architecture specialized to run calculations only on the non-zero values in the matrices. The resulting chip design is therefore tailored to maximize efficiency for the robot’s computing needs. And that customization paid off in testing.

Hardware architecture designed using this method for a particular application outperformed off-the-shelf CPU and GPU units. While Neuman’s team didn’t fabricate a specialized chip from scratch, they programmed a customizable field-programmable gate array (FPGA) chip according to

their system's suggestions. Despite operating at a slower **clock rate**, that chip performed eight times faster than the CPU and 86 times faster than the GPU.

"I was thrilled with those results," says Neuman. "Even though we were **hamstrung** by the lower clock speed, we made up for it by just being more efficient."

Plancher sees widespread potential for robomorphic computing. "Ideally we can eventually fabricate a **custom motion-planning** chip for every robot, allowing them to quickly compute safe and efficient motions," he says. "I wouldn't be surprised if 20 years from now every robot had a handful of custom computer chips powering it, and this could be one of them." Neuman adds that robomorphic computing might allow robots to relieve humans of risk in a range of settings, such as caring for covid-19 patients or manipulating heavy objects.

"This work is exciting because it shows how specialized circuit designs can be used to accelerate a core component of robot control," says Robin Deits, a robotics engineer at Boston Dynamics who was not involved in the research. "Software performance is crucial for robotics because the real world never waits around for the robot to finish thinking." He adds that Neuman's advance could enable robots to think faster, "**unlocking** exciting behaviors that previously would be too computationally difficult."

Neuman next plans to automate the entire system of robomorphic computing. Users will simply **drag and drop** their robot's parameters, and "out the other end comes the hardware description. I think that's the thing that'll push it over the edge and make it really useful."

This research was funded by the National Science Foundation, the Computing Research Agency, the **CIFellows Project**, and the Defense Advanced Research Projects Agency.

The material is available at <https://news.mit.edu/2021/robot-customized-hardware-0121>

Words and expressions:

1. hang up: 搁置; 悬挂

2. stimuli: 刺激

3. boatload: 一条船的货物; 大量

4. robomorphic: 机器人形态的

5. contagious: 传染的

6. fellow: 研究员

7. algorithms ['ælgərɪðəm]: 算法

8. standard-issue: [军]常规的

9. graphics processing unit, GPU: 图形处理器

10. handy: 方便的; 手边的

11. pixels: 像素

12. sparse: 稀少的; 稀疏的

13. pliable: 易曲折的; 柔软的

14. tailored: 特制的; 合身的

15. paid off: 付清; 得到好处/回报

16. off-the-shelf: 成品的; 现成的

17. gate array: 门数组, 门阵列

18. clock rate: 时钟速率

19. hamstring: 残废的; 做不了的 (hamstring 的过去式和过去分词形式)

20. custom motion-planning: 定做的/定制的移动规划

21. unlock: 开锁; 开启; 揭开

22. drag and drop: 拖放

23. CIFellows Project: Computing Innovation Fellows Project, 计算创新研究员项目

Reading comprehension questions

1. What slows down the response time of robots in complex situations?
2. What is the way Neuman has found to fight the mismatch between a robot's "mind" and body?
3. What are major steps in a robot's operation?
4. What could be widespread potential for robomorphic computing?

Unit 5

第一篇:

3D Printing Polymers

The material **yields** 【产生】 soft, elastic objects that feel like human tissue

Researchers in the labs of Christopher Bates, an assistant professor of materials at UC Santa Barbara, and Michael Chabinyc, a professor of materials and chair of the department, have teamed to develop the first 3D-printable "bottlebrush" **elastomer**^[1]. The new material results in printed objects that have unusual softness and elasticity -- mechanical properties that closely resemble those of human tissue.

Conventional elastomers, i.e. rubbers, are stiffer than many biological tissues. That's due to the size and shape of their constituent **polymers**^[2], which are long, linear molecules that easily entangle like cooked spaghetti. In contrast, bottlebrush polymers have additional polymers attached to the linear backbone, leading to a structure more **akin to** 【与...类似】 a bottle brush you might find in your kitchen. The bottlebrush polymer structure **imparts** 【赋予, 给予〔某种品质〕】 the ability to form extremely soft elastomers.

The ability to 3D-print bottlebrush elastomers makes it possible to leverage these unique mechanical properties in applications that require careful control over the dimensions of objects

^[1] n. [力] 弹性体, [高分子] 高弹体

^[2] n. [高分子] 聚合物

ranging from **biomimetic**^[3] tissue to high-sensitivity electronic devices, such as touch pads, sensors and actuators.

Two postdoctoral researchers -- Renxuan Xie and Sanjoy Mukherjee -- played key roles in developing the new material. Their findings were published in the journal *Science Advances*.

Xie's and Mukherjee's key discovery involves the self-assembly of bottlebrush polymers at the nanometer length scale, which causes a solid-to-liquid transition **in response to** **【响应；回答；对...做出反应】** applied pressure. This material is categorized as a yield-stress fluid^[4], meaning it begins as a semi-soft solid that **holds its shape**, like butter or toothpaste, but when sufficient pressure is applied, it liquefies and can be squeezed through a syringe. The team exploits this property to create inks in a 3D-printing process called direct ink writing (DIW).

The researchers can **tune** **【调整】** the material to flow under various amounts of pressure to match the desired processing conditions. "For instance, maybe you want the polymer to hold its shape under a different level of stress, such as when vibration is present," says Xie. "Our material can hold its shape for hours. That's important, because if the material **sags**^[5] **【下凹】** during printing, the printed part will have poor structural stability."

^[3] adj. 仿生的

^[4] 屈服应力流体

^[5] v. 中间下垂，下凹

Once the object is printed, UV light is shined onto it to activate crosslinkers^[6] that Mukherjee synthesized and included as a part of the ink formulation. The crosslinkers can link up 【连接】 nearby bottlebrush polymers, resulting in a super-soft elastomer. At that point, the material becomes a permanent solid -- it will no longer liquefy^[7] under pressure -- and exhibits extraordinary properties.

"We start with long polymers that are not crosslinked," said Xie. "That allows them to flow like a fluid. But, after you shine the light on them, the small molecules between the polymer chains react and are linked together into a network, so you have a solid, an elastomer that, when stretched, will return to its original shape."

The softness of a material is measured in terms of its modulus^[8], and for most elastomers, it is rather high, meaning their stiffness and elasticity are similar to those of a rubber band. "The modulus of our material is a thousand times smaller than that of a rubber band," Xie notes. "It is super-soft -- it feels very much like human tissue -- and very stretchy. It can stretch about three to four times its length."

An Accidental Ink (how this material is developed)

Mukherjee discovered the material by accident, while trying to develop a material for a different project, one that would increase the amount of charge that can be stored by an actuator^[9]. When

^[6] 固化剂

^[7] vt. 液化; 溶解

^[8] n. 系数

^[9] 传动装置

the elastomer came to Xie for characterization, he knew immediately that it was special. "I could see right away that it was different, because it could hold its shape so well," he recalled.

"When we saw this really well-defined yield stress, it dawned on everyone collectively that we could 3D-print it," Bates said, "and that would be cool, because none of the 3D-printable materials we know of have this super-soft property."

Bottlebrush polymers have been around for more than twenty years. But, Bates said, "The field has exploded in the past ten years thanks to advances in synthetic chemistry that provide exquisite control over the size and shape of these unique molecules.

"These super-soft elastomers might be applicable as [implants^{\[10\]}](#)," he added. "You may be able to reduce inflammation and rejection by the body if the mechanical properties of an implant match native tissue."

Another important element of the new material is that it is pure polymer, Chabinyc noted.

"There's no water or other solvent in them to artificially make them softer," he said.

To understand the importance of having no water in the polymer, it's helpful to think of Jell-O, which is mostly water and can hold its shape, but only as long as the water remains inside. "If the water went away, then you'd just have a shapeless pile of material," said Chabinyc. "With a

^[10] 移植

conventional polymer, you must figure out how to keep the right amount of water in it to maintain its structure, but this new material is all solid, so it will never change."

Moreover, the new material can be 3D-printed and processed without solvent, which is also unusual. "People often add solvent to liquify a solid so that it can be squeezed out of a nozzle," said Xie, "but if you add solvent, it has to evaporate after printing causing the object to change its shape or crack."

Mukherjee added, "We wanted the material and the printing process to be as clean and as easy as possible, so we played a chemistry trick with solubility and self-assembly, which enabled the solvent-free process. The fact that we don't use solvent is a tremendous advantage."

Story Source:

[Materials](#) provided by [University of California - Santa Barbara](#). Original written by James Badham. Note: Content may be edited for style and length.

Journal Reference:

1. Renxuan Xie, Sanjoy Mukherjee, Adam E. Levi, Veronica G. Reynolds, Hengbin Wang, Michael L. Chabinyc, Christopher M. Bates. **Room temperature 3D printing of super-soft and solvent-free elastomers**. Science Advances, 2020; 6 (46): eabc6900 DOI: [10.1126/sciadv.abc6900](https://doi.org/10.1126/sciadv.abc6900)

CHEMISTRY

Chemists are reimagining recycling to keep plastics out of landfills

Too much of today's plastic is impossible to recycle



A lot of plastic goes to landfills because the material is too difficult to recycle into useful new products.

ABDUL RAHEEM MOHAMED/EYEEM/GETTY IMAGES

By **Maria Temming**

JANUARY 27, 2021 AT 10:19 AM

It feels good to recycle. There's a certain sense of accomplishment that comes from dutifully sorting soda bottles, plastic bags and yogurt cups from the rest of the garbage. The more plastic you put in that blue bin, the more you're keeping out of landfills and the oceans, right?

Wrong. No matter how meticulous you are in cleaning and separating your plastics, most **end up** 【最终成为; 最后处于】 in the trash heap anyway.

Take flexible food packages. Those films contain several layers of different plastics. Because each plastic has to be recycled separately, those films are not recyclable. Grocery bags and shrink wrap are too flimsy, **prone to 【易于 ... 的】 getting tangled up with** other materials on a conveyor belt. The **polypropylene**^[11] in yogurt cups and other items doesn't usually get recycled either; recycling a **hodgepodge**^[12] of polypropylene produces a dark, smelly plastic that few manufacturers will use.

Only two kinds of plastic are commonly recycled in the United States: the kind in plastic soda bottles, **polyethylene terephthalate** ^[13], or **PET**^[14]; and the plastic found in milk jugs and detergent containers — high-density polyethylene, or HDPE.

Together, **those plastics make up only about a quarter of the world's plastic trash**, researchers reported in 2017 in *Science Advances*. And when those plastics are recycled, they aren't good for much. Melting plastic down to recycle changes its consistency, so PET from bottles has to be mixed with brand-new plastic to make a sturdy final product. **Recycling a mix of multicolored HDPE pieces creates a dark plastic good only for making products like park benches and waste bins, in which properties like color don't matter much.**

The difficulties of recycling plastic into anything manufacturers want to use is a big reason why the world **is littered with 【充满某物】** so much plastic waste, says Eric Beckman, a chemical engineer at the University of Pittsburgh. In 2018 alone, the United States **landfilled 27 million tons of plastic** and

[11] [ˌpɒliˈprɒpəliːn] [高分子] 聚丙烯

[12] [ˈhɒdʒpɑːdʒ] n. 烩; 混煮; 一团糟

[13] [ˌtɛrɪfˈθæleɪtɪn] 对苯二酸盐; 对苯二酸酯

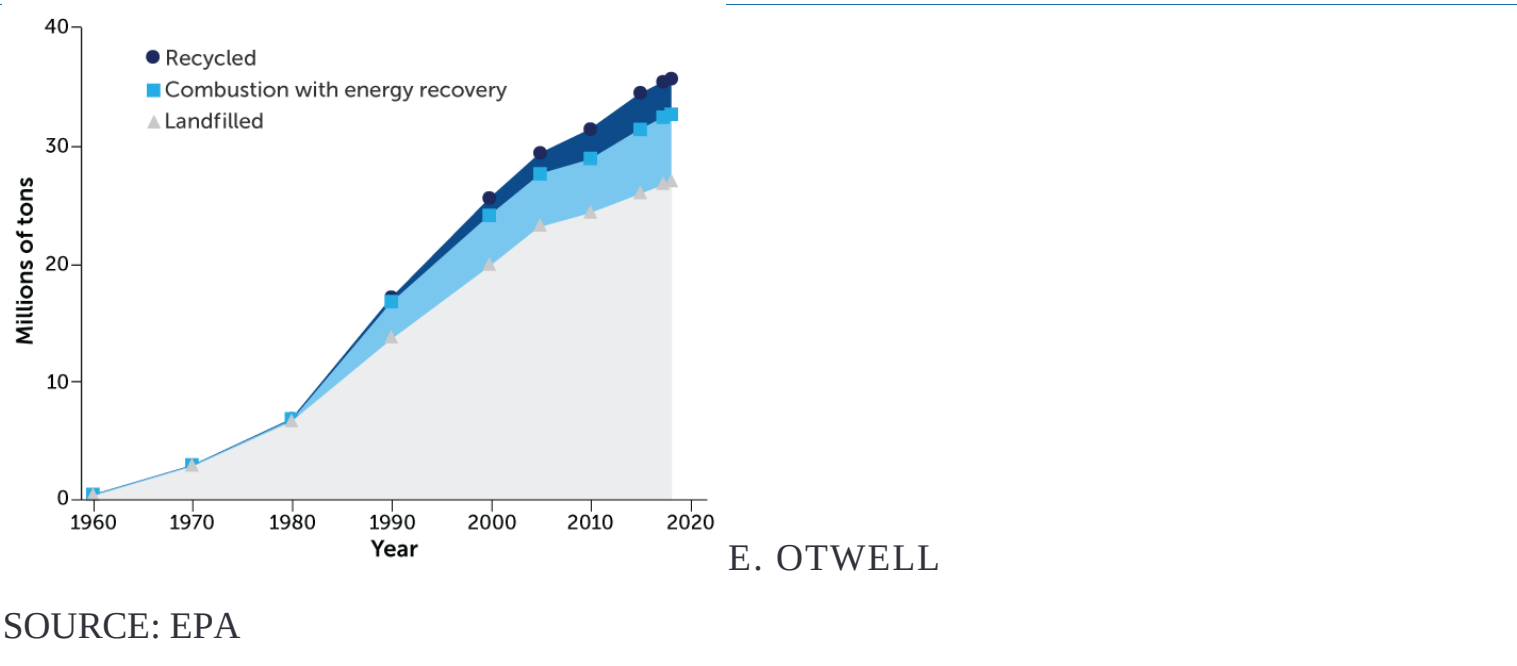
[14] 聚对苯二甲酸乙二醇酯

recycled a mere 3 million, according to the U.S. Environmental Protection Agency. Low recycling rates aren't just a problem in the United States. Of the 6.3 billion tons of plastic that have been discarded around the world, only about 9 percent has gotten recycled. Another 12 percent has been burned, and almost 80 percent has **piled up**【堆积】 on land or in waterways.

Good news/bad news

The amount of plastic recycled in the United States has increased over the last few decades — but those levels still **pale**【苍白的】 in comparison with the amount of plastic that goes into landfills.

Plastic waste management, 1960–2018



With plastic collecting everywhere from the top of **Mount Everest**^[15] to the bottom of the **Mariana Trench**^[16], there's an **urgent need to reduce the amount of plastic** that gets thrown away (SN: 1/16/21,

^[15] 埃佛勒斯峰, 喜马拉雅山主峰之一, 中国称珠穆朗玛峰
^[16] 马里亚纳海沟 (太平洋海底)

p. 5). Some people propose replacing plastics with biodegradable materials, but those replacements are generally not as strong or cheap to make as plastics (SN: 6/22/19, p. 18). Since, realistically, plastic is not going away any time soon, chemists who understand the ins and outs of all this pesky^[17] plastic are working to make it easier to recycle and turn into higher-quality material that's useful for more things.

“There's not going to be a single technology that's going to be the answer,” says Ed Daniels, senior project manager at the REMADE Institute in West Henrietta, N.Y., which funds research into new recycling techniques. Some projects are on the brink of 【濒临；在.....的边缘】 breaking into industry; others are still just promising lab experiments. But all are focused on designing a future where any plastic that ends up in the recycling bin can have a second and third life in a new product.

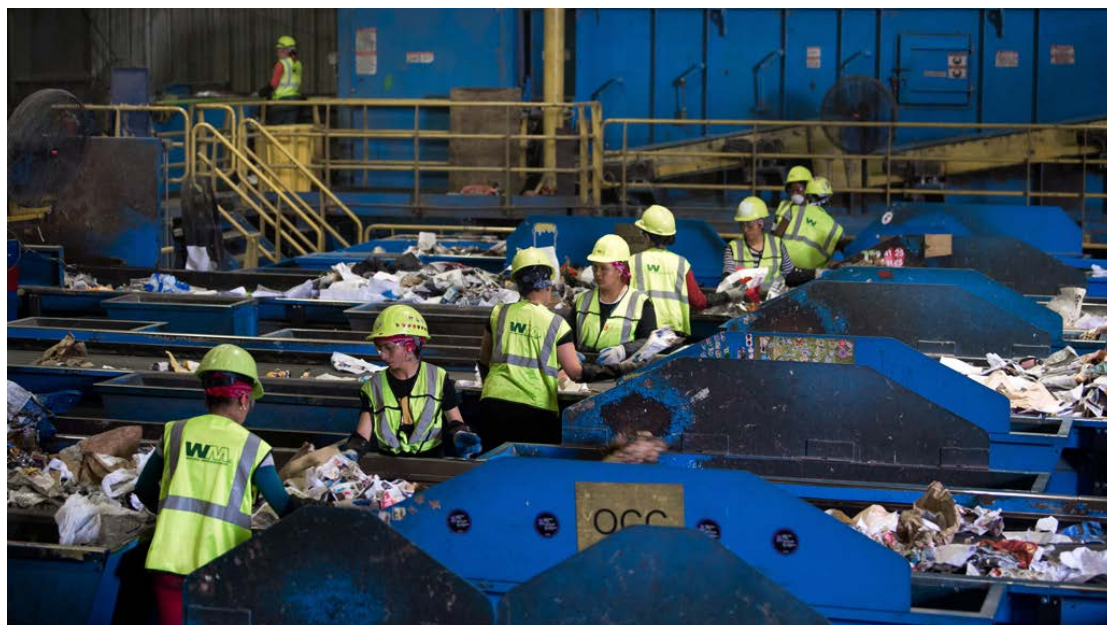
Picking plastics apart

One of the biggest bottlenecks in plastic recycling is that every material has to get processed separately. “Most plastics are like oil and water,” says chemist Geoffrey Coates of Cornell University. They just don't mix. Take, for example, a polyethylene detergent jug and its polypropylene cap. “If you melt those down, and I make a bottle out of that, and I squeeze it, it would basically crack down the side,” Coates says. “It's crazy brittle. Totally worthless.”

That's why the first destination for plastic recyclables is a material recovery facility, where people and machines do the sorting. Separated plastics can then be washed, shredded, melted and remolded. The

^[17] adj. 讨厌的，麻烦的

system works well for simple items like soda bottles and milk jugs. But not for items like **deodorant** ^[18]containers — where the bottle, crank and cap could all be made of different kinds of plastic. Food packaging films that contain several layers of different plastic are particularly tricky to take apart. Every year, 100 million tons of these multilayer films are produced worldwide. When thrown away, those plastics go to landfills, says chemical engineer George Huber of the University of Wisconsin–Madison.



At the Waste Management Material Recovery Facility in Elkridge, Md., workers sort trash moving past them on conveyor belts. SAUL LOEB/AFP VIA GETTY IMAGES

To tackle that problem, Huber and colleagues devised a strategy for dealing with complex mixtures of plastics. The process uses a series of liquid solvents to **dissolve** 【溶解】 individual plastic

^[18] adj. 除臭的，防臭的

components off a product. The trick is choosing the right solvents to dissolve only one kind of plastic at a time, Huber says.

The team tested the technique on a packaging film that contained polyethylene and PET, as well as a plastic oxygen barrier made of [ethylene vinyl alcohol](#)^[19], or EVOH, that keeps food fresh.

Stirring the film into a [toluene](#)^[20] solvent first dissolved the polyethylene layer. Dunking the remaining EVOH-PET film in a solvent called [DMSO](#)^[21] stripped off the EVOH. The researchers then [plucked out](#) [【拔掉, 褪去】](#) the remaining PET film and recovered the other two plastics from their separate solvents by mixing in “antisolvent” chemicals. Those chemicals caused the plastic molecules that were [dispersed](#) [【分散】](#) in the liquids to bunch together into solid clumps that could be [fished out](#) [【\(从水中\)吊起; 捞出】](#).

This process [recovered practically all of the plastic from the original film](#), the researchers reported last November in *Science Advances*. When tested on a jumble of polyethylene, PET and EVOH beads, the solvent washes recovered more than 95 percent of each material — hinting that these solvents could be used to strip plastic components off bulkier items than packaging films. So in theory, recovery facilities could use this technique to disassemble multiplastic deodorant containers and other products of various shapes and sizes.

[19] 乙烯乙烯醇

[20] 甲苯

[21] 二甲基亚砷

Huber and colleagues next plan to look for solvents to dissolve more kinds of plastic, such as the polystyrene [22]in Styrofoam[23] . But it will take a lot more work to make this strategy efficient at sorting all the intricate plastic combinations in real-world recyclables.

False advertising

Many plastic products are labeled with a number inside a triangle that symbolizes recycling. Yet, only plastics with 1 (polyethylene terephthalate) or 2 (high-density polyethylene) are widely recycled in the United States. The rest typically go to the landfill.



Water and soft drink bottles, salad domes, cookie trays, salad dressing and peanut butter containers



Milk and juice bottles, freezer bags, shampoo and detergent bottles

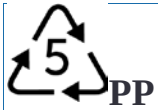


Cosmetic containers, commercial cling wrap 保鲜膜

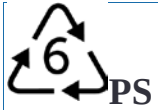


Squeeze bottles, cling wrap, trash bags

[22] 聚苯乙烯
[23] 泡沫聚苯乙烯



Microwave dishes, ice cream tubs, yogurt containers, detergent bottle caps



CD cases, plastic disposable cups, plastic cutlery, video cases



Foam polystyrene hot drink cups, food takeaway trays, protective packaging for fragile items



Water cooler bottles, flexible films, multimaterial packaging

SOURCE: ELLEN MACARTHUR FOUNDATION 2017

Making plastics mix

There may also be chemical shortcuts that allow multilayer films and other mixtures of plastics to be recycled as they are. Additives called **compatibilizers**^[24] help different melted-down plastics blend, so that unsorted materials can be treated as one. But there is no universal compatibilizer that allows every kind of plastic to be mixed together. And existing compatibilizers are not widely used because they are not very **potent**【有效的；强有力的】—— and adding a lot of compatibilizer to a plastic blend gets expensive.

^[24] n. [助剂] 增容剂

To boost viability, Coates and colleagues created a highly potent compatibilizer for polyethylene and polypropylene. Together, those two plastics make up more than half of the world's plastic. The new compatibilizer molecule contains two segments of polyethylene, interspersed with two segments of polypropylene. Those alternating segments latch onto plastic molecules of the same kind in a mixture, bringing polyethylene and polypropylene together. It's as if polyethylene were made of Legos, and polypropylene were made of Duplos, and the researchers made a special building block with connectors that fit both types of blocks.

Having two polyethylene and two polypropylene connectors for each compatibilizer molecule, rather than one, [made this compatibilizer stronger than previous versions](#), Coates and colleagues reported in 2017 in *Science*. The first test of the new compatibilizer involved welding together strips of polyethylene and polypropylene. Ordinarily, the two materials easily peel apart. But with a layer of compatibilizer between them, the plastic strips broke, rather than the compatibilizer seal, when pulled apart.

In a second test, the researchers mixed the compatibilizer into a melted blend of polyethylene and polypropylene. It took only 1 percent compatibilizer to create a tough new plastic.

"These are crazy potent additives," Coates says. Other compatibilizers had to be added at concentrations up to 10 percent to hold these two plastics together. The new compatibilizer is now the basis for Coates' start-up, Intermix Performance Materials, based in Ithaca, N.Y.

Good as new

Even if every piece of plastic trash could easily be recycled, that still wouldn't solve the world's plastic problem. There are a couple major issues with how recycling currently works that severely limit the usability of recycled materials.

For one thing, recycled plastics inherit all the dyes, flame retardants and other additives that gave each original plastic piece its distinctive look and feel. “The plastic that you actually recover at the end of all this is really a very complex mixture,” says chemist Susannah Scott of the University of California, Santa Barbara. Few manufacturers can use plastic with a random **mishmash** ^[25] of properties to make something new.

Plus, recycling breaks some of the chemical bonds in plastic molecules, affecting the strength and consistency of the material. Melting down and remolding plastic is sort of like reheating pizza in the microwave — you get out basically what you put in, just not as good. That limits the number of times plastic can be recycled before it has to be landfilled.

The solution to both problems could lie in a new kind of recycling process, called chemical recycling, which promises to make pure new plastic an infinite number of times. Chemical recycling involves taking plastics apart on the molecular level.

[25] n. 混合物

The molecules that make up plastics are called polymers, which are made of smaller [monomers](#)^[26]. Using heat and chemicals, it is possible to [disassemble](#) [【拆卸】](#) polymers [into](#) monomers, separate those building blocks from dyes and other contaminants, and piece the monomers back together into good-as-new plastic.

“Chemical recycling has really started to emerge as a force, I would say, within the last three or four years,” says University of Pittsburgh’s Beckman. But most chemical recycling techniques are too expensive or energy intensive for commercial use. “It’s not ready for prime time,” he says.

Different plastics require different chemical recycling processes, and some break down more easily than others. “The one that’s farthest along is PET,” Beckman says. “That polymer happens to be easy to take apart.” Several companies are developing methods to chemically recycle PET, including the French company Carbios.

Carbios is testing enzymes produced by microorganisms to break down PET. Researchers at the company [described their work](#) on one such enzyme last April in *Nature*. Microbes normally use the enzyme, called leaf-branch [compost](#)^[27] [cutinase](#)^[28], to decompose the waxy coating on plant leaves. But the cutinase is also good at breaking PET down into its monomers: [ethylene glycol](#)^[29] and [terephthalic acid](#)^[30].

[26] 单体

[27] 堆肥：混合物

[28] 角质酶

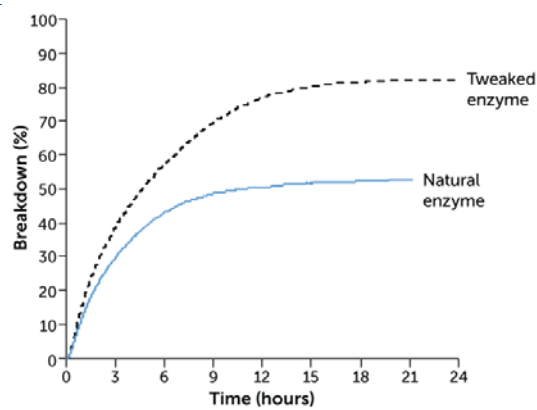
[29] 乙二醇

[30] 对苯二甲酸

Microbial help

An enzyme naturally produced by microbes broke down about 50 percent of polyethylene terephthalate, or PET (blue line). A tweaked version of the enzyme broke down more than 80 percent of the plastic (black dotted line). Increasing the amount of the enzyme from 1 milligram per gram of PET to 3 milligrams made it even more efficient — breaking down about 90 percent of PET.

PET breakdown by an enzyme



E. OTWELL

SOURCE: V. TOURNIER *ET AL*/NATURE 2020

“The enzyme is like a molecular scissor,” says Alain Marty, chief scientific officer at Carbios. But because it evolved to decompose plant matter, not plastic, it’s not perfect. To make the enzyme better at snipping apart PET, “we redesigned what we call the active site of the enzyme,” Marty says. This involved swapping out some of the amino acids^[31] along that PET docking site for others.

When the researchers tested their mutant enzyme on colored plastic flakes from PET bottles, applying 3 milligrams of the enzyme per gram of PET, about 90 percent of the plastic broke down in

^[31] [生化] 氨基酸

about 10 hours. The original enzyme had maxed out at about 50 percent. Using the terephthalic acid monomers produced in that process, the researchers made new plastic bottles that were just as strong as the originals.

Carbios is now building a plant near Lyon, France, to start chemically recycling PET later this year.

Milder conditions

But other plastics, like polyethylene and polypropylene, are much harder to break down via chemical recycling. Taking apart polyethylene molecules, for instance, requires temperatures over 400° Celsius. At such high heat, the chemistry is chaotic. Plastic molecules break down randomly, generating a complex mixture of compounds that can be burned as fuel but not used to make new materials.

Scott, the UC Santa Barbara chemist, proposes partially breaking down these sturdy plastics in a more controlled way, under milder conditions, to make other kinds of useful molecules. She and colleagues recently came up with a way to transform polyethylene into alkylaromatic^[32] compounds, which can be used as biodegradable ingredients in shampoos, detergents and other products. The process involves placing polyethylene inside a reaction chamber set to 280° C, with a catalyst powder containing platinum nanoparticles^[33].

Polyethylene is a long molecule, in which hydrogen atoms are connected to a carbon backbone that can be thousands of carbon atoms long. The platinum is good at breaking carbon-hydrogen bonds, Scott says. “When you do that, you generate hydrogen in the reactor, and the platinum catalyst can

^[32] 烷基芳烃的

^[33] 铂纳米颗粒

use the hydrogen to break the carbon-carbon bonds [in the molecule backbone]. So it actually chops the chain into smaller pieces.”

Since this reaction takes place at a relatively mild 280° C, it happens in an orderly fashion, snapping long polyethylene molecules into shorter chains that are each about 30 carbons long. Those fragments then arrange themselves into the six-sided ring structures characteristic of alkylaromatic compounds.

After 24 hours in the reaction chamber, “most of the products are liquids, and most of the liquids are alkylaromatics,” Scott says. In experiments, about 69 percent of the plastic in a low-density polyethylene bag [was converted into liquid](#). About 55 percent of a high-density polyethylene bottle cap was transformed. The process produces hydrocarbon gases too, which could be used to generate heat to run the reaction at a recycling plant, Scott says.

For now, this is just a lab demo, and like many new recycling strategies, it’s still a long way off from commercialization. And no single upgrade to the recycling pipeline will rid the world of its growing mountains of plastic trash. “We’re going to need a suite of technologies to meet this challenge,” says Daniels, of the REMADE Institute. But each new technology — whether it’s focused on making plastics easier to recycle, or transforming them into more useful materials — could help.

Built to last

The plastics produced today were never designed to be used more than once. That's why recycling plastics — particularly into material that is as good as new — is so difficult. But researchers are going back to the drawing board to ask themselves, “What does the next generation of materials look like? How do you design a material specifically so that it never has to go into a landfill?” says Eric Beckman, a chemical engineer at the University of Pittsburgh. “Chemists are looking at whether you can design a polymer that falls apart on command.”

The development of a class of next-gen polymers, called PDKs, for poly(diketoenamine)s^[34], was reported in *Nature Chemistry* in 2019. “PDKs have the ability to break their bonds under relatively mild conditions — certainly with much lower energy intensity than any of the plastics that are currently used today,” says study coauthor Brett Helms, a chemist at the Lawrence Berkeley National Laboratory in California. Simply dunking the plastic in an acid solution with a pH of 1 or 2 is enough to break the bonds between its monomer building blocks.

“Materials don't usually encounter a pH that's that low, so it's not like if you put PDKs in vinegar, the polymer is going to start breaking down,” Helms says. But it could make for easy recycling. The PDK monomers can then be used to make pristine new plastic, again and again.

Widespread plastics like polyethylene terephthalate, or PET, and polyethylene are so cheap to make that any breakout polymer would have a tough time entering the market, Beckman says. So for now, inherently recyclable plastic is just an academic curiosity. But maybe, decades from now, plastics

^[34] 聚二酮烯胺

made to be recyclable from the get-go will help solve the world's plastic waste problem. — *Maria Temming*